

D-RECIPE: Dynamic Continual Learning for Temporal Knowledge Graph Forecasting via Incremental Rule Mining and LLM Adaptation

Busisani Mac Dube¹ Jean Vincent Fonou Dombeu¹

¹Department of Computer Science, North-West University, South Africa

{busisani.macdube, jvfou}@nwu.ac.za

Draft — March 17, 2026 [arXiv pre-print]

Abstract

“The dynamic nature of knowledge poses a challenge, as KGs must continuously evolve to reflect changes in the world. Most existing work assumes static KGs, neglecting the continuous addition of new edges and the modification of existing relationships” [Mac Dube and Fonou Dombeu, 2023]. Temporal Knowledge Graph (TKG) forecasting—predicting future facts given a history of time-stamped events—has seen remarkable progress with rule-based and large language model (LLM)-based approaches. However, virtually all existing methods, including the recent state-of-the-art RECIPE-TKG [Akgul et al., 2025], treat the KG as a static snapshot: they require full re-training and full rule re-mining when the graph changes, making them fundamentally incompatible with real-world deployment scenarios where edges arrive continuously as a stream.

We introduce D-RECIPE (**D**ynamic **R**ule-bas**E**d **C**ontinual **P**redictor), the first LLM-based continual TKG completion framework that handles streaming edge ingestion without offline reprocessing. D-RECIPE extends RECIPE-TKG with five novel components: (1) an *incremental rule miner* that updates temporal rules in $O(k)$ per new edge without full re-mining; (2) an *edge stream processor* with $O(1)$ adjacency updates and emergence detection; (3) a *continual learning framework* combining Elastic Weight Consolidation (EWC), knowledge distillation (KD), and experience replay to prevent catastrophic forgetting; (4) an *adaptive Rule-Based Multi-Hop (RBMH) sampler* with incremental hop-cache invalidation; and (5) full Apple M-series (MPS) hardware compatibility. Evaluated across four standard TKG benchmarks (ICEWS14, ICEWS18, GDELT, YAGO), D-RECIPE achieves Hits@10 of 0.681, 0.543, 0.349, and 0.945, surpassing RECIPE-TKG by up to +8.4% in the streaming setting. Continual learning metrics confirm near-zero backward transfer (BWT = +0.005, ICEWS14),

demonstrating that temporal knowledge is preserved as the graph evolves.

Keywords: Temporal Knowledge Graphs, Continual Learning, LLM Fine-tuning, Rule Mining, Knowledge Graph Completion, Catastrophic Forgetting, Streaming Graphs.

1 Introduction

Knowledge Graphs (KGs) encode world knowledge as triples (s, r, o) —subject, relation, object—enabling downstream reasoning, question answering, and recommendation [Hogan et al., 2021]. *Temporal* KGs (TKGs) extend this representation to quadruples (s, r, o, t) capturing events timestamped with a discrete time step t . TKG *forecasting*—predicting future facts from historical context—is a practically important task underpinning event prediction in geopolitical analysis (ICEWS, GDELT), encyclopaedic knowledge maintenance (YAGO), and scientific knowledge evolution.

Despite rapid progress [Li et al., 2021, Sun et al., 2021, Liao et al., 2024, Akgul et al., 2025], the field operates under a critical assumption: *the graph is static*. Rule mining is performed once on the full training set; LLMs are fine-tuned on a fixed corpus; evaluation is done on a snapshot. Yet real TKGs are inherently dynamic—new entities emerge daily, relations are asserted and retracted, and temporal patterns shift. As noted by Mac Dube and Fonou Dombeu [2023], this static assumption is a fundamental gap in the literature. The naive solution—rerun everything when the graph changes—is computationally prohibitive. Full TLogic rule mining [Liu et al., 2022b] on ICEWS14 takes ≈ 8 minutes per update; full fine-tuning of a 7B-parameter LLM takes hours.

This paper closes the gap with D-RECIPE: a framework that makes LLM-based TKG forecasting com-

patible with streaming graphs via principled continual learning. D-RECIPE is built on RECIPE-TKG [Akgul et al., 2025] and retains all of its inductive components while adding the machinery needed for streaming operation.

Main Contributions.

1. **Incremental Rule Mining.** We derive an $O(k)$ update rule for temporal rule confidence that eliminates full re-mining when new edges arrive (section 4.2).
2. **Edge Stream Processor.** An $O(1)$ adjacency-list-based stream processor that tracks entity/relation emergence and supports time-bucketed snapshot queries (section 4.1).
3. **Continual Learning Framework.** A multi-objective loss $\mathcal{L} = \mathcal{L}_{CE} + \alpha\mathcal{L}_C + \beta\mathcal{L}_{EWC} + \gamma\mathcal{L}_{KD}$ combining cross-entropy, contrastive, EWC, and knowledge distillation terms with reservoir-sampled experience replay to prevent catastrophic forgetting (section 4.4).
4. **Adaptive RBMH Sampler.** Dynamic hop-distance recalculation with incremental cache invalidation for the subgraph affected by each new edge, retaining RECIPE-TKG’s rule-guided composite weights (section 4.3).
5. **Full Apple Silicon (MPS) Compatibility.** Device-agnostic BF16 precision training, replacing CUDA-only bitsandbytes quantisation, enabling experiments on M-series Macs without modification (section 5.4).

2 Related Work

2.1 Temporal Knowledge Graph Completion

Early TKG methods such as RE-GCN [Li et al., 2021] and xERTE [Han et al., 2021a] model entity evolution with graph neural networks and temporal attention, respectively. TANGO [Han et al., 2021b] uses neural ordinary differential equations to continuously model temporal graph dynamics; however, all of these methods treat the training-time graph as fixed. TimeTraveler [Sun et al., 2021] introduces reinforcement learning for search-path reasoning over snapshots. TLogic [Liu et al., 2022b] mines temporal Horn rules and uses them for explicit multi-step reasoning, but rule mining scales quadratically with graph size and must be re-run on any change.

2.2 LLM-Based TKG Forecasting

In-Context Learning (ICL) approaches [Xiong et al., 2024] prompt frozen LLMs with historical facts as context; GenTKG [Liao et al., 2024] combines rule retrieval with in-context few-shot prompting. RECIPE-TKG [Akgul et al., 2025] achieves state-of-the-art results by combining (i) Rule-Based Multi-Hop retrieval (RBMH) for context sampling, (ii) contrastive LoRA fine-tuning, and (iii) semantic test-time filtering. Its explicit limitation—“Rule mining must be rerun if the graph changes”—directly motivates this work.

2.3 Continual Learning

Catastrophic forgetting [McCloskey and Cohen, 1989] is the central challenge in continual learning (CL). Elastic Weight Consolidation (EWC) [Kirkpatrick et al., 2017] penalises changes to parameters that are important for previous tasks, quantified via the Fisher information matrix. Knowledge Distillation [Hinton et al., 2015] preserves soft predictions of a frozen teacher as new tasks are learned. Experience Replay [Robins, 1995] mitigates forgetting by interleaving past examples with new ones. Our framework combines all three mechanisms, adapting them to the structured-prediction setting of TKG forecasting. Recent work on continual KG embedding [Liu et al., 2022a] focuses on static triple completion; we are, to our knowledge, the first to apply CL to LLM-based *temporal* link prediction.

2.4 Streaming / Incremental KG Methods

StreamE [Luo et al., 2023] proposes streaming embeddings for KGs but does not handle temporal reasoning rules or LLM integration. Incremental rule learning for static KGs has been studied in the ILP community [Galárraga et al., 2013]; we adapt this idea to *temporal* confidence updates under a streaming model.

3 Preliminaries and Problem Formulation

Definition 1 (Temporal Knowledge Graph). A *Temporal Knowledge Graph* is a set $\mathcal{G} = \{(s, r, o, t) \mid s, o \in \mathcal{E}, r \in \mathcal{R}, t \in \mathcal{T}\}$ where $\mathcal{E}$ is the entity set, $\mathcal{R}$ the relation set, and $\mathcal{T} \subset \mathbb{Z}^+$ discrete timestamps.

Definition 2 (Temporal Link Prediction). Given a query $(s_q, r_q, ?, t_q)$ with $t_q > \max \mathcal{T}$, rank all entities $e \in \mathcal{E}$ by $P(o = e \mid s_q, r_q, t_q, \mathcal{G})$. Performance is

measured by Hits@ k (fraction of queries where the true object is in the top- k predictions) and MRR.

Definition 3 (Streaming TKG). A *streaming TKG* partitions facts into ordered chunks $\mathcal{G} = \mathcal{G}_1 \cup \mathcal{G}_2 \cup \dots \cup \mathcal{G}_N$, where $\max_{(s,r,o,t) \in \mathcal{G}_i} t < \min_{(s,r,o,t) \in \mathcal{G}_{i+1}} t$. At period i , only $\mathcal{G}_1 \cup \dots \cup \mathcal{G}_i$ is observed; predictions are made for queries at $t > \max_i$.

Definition 4 (Continual TKG Forecasting). A *continual TKG forecasting* system must maintain a predictor f_θ that, at each period i , (a) ingests $\mathcal{G}_i$ online, (b) updates θ without full reprocessing of $\mathcal{G}_{<i}$, and (c) improves (or does not degrade) Hits@ k on queries at $t > i$.

Continual Learning Metrics. Following Lopez-Paz and Ranzato [2017], we measure:

$$\text{BWT} = \frac{1}{T-1} \sum_{i=1}^{T-1} (A_{T,i} - A_{i,i}), \quad (1)$$

$$\text{FWT} = \frac{1}{T-1} \sum_{i=2}^T (A_{i-1,i} - \hat{b}_i), \quad (2)$$

$$\text{AvgAcc} = \frac{1}{T} \sum_{i=1}^T A_{T,i}, \quad (3)$$

where $A_{j,i}$ is Hits@10 evaluated on period i after training on period j , and $\hat{b}_i$ is a random-initialisation baseline for period i . $\text{BWT} \approx 0$ (no forgetting); $\text{BWT} < 0$ (catastrophic forgetting); $\text{FWT} > 0$ (positive forward transfer).

4 D-RECIPE Architecture

Figure 1 illustrates the full D-RECIPE pipeline. At each streaming step, incoming edges $\{(s, r, o, t)\}$ flow through five sequential components before being used to update the LLM-based predictor.

4.1 Edge Stream Processor

The stream processor maintains dynamic adjacency lists:

$$\text{adj}_{\text{out}}[s] += (r, o, t), \quad \text{adj}_{\text{in}}[o] += (r, s, t), \quad (4)$$

and a time-indexed bucket map $\mathcal{B}[t] += (s, r, o, t)$. Each ingestion is $O(1)$ in hash-table time. The processor detects *new entities* (absent from $\mathcal{E}$ before this chunk) and *new relations* (absent from $\mathcal{R}$), triggering partial re-routing in the sampler.

4.2 Incremental Rule Mining

Temporal rules take the form:

$$r_1(X, Y, t_1) \wedge r_2(Y, Z, t_2) \Rightarrow r_h(X, Z, t_3), \quad t_1 < t_2 < t_3. \quad (5)$$

In the static setting (TLogic), the confidence of such a rule is:

$$\text{conf}(\pi) = \frac{|\{(X, Z, t_3) \text{ satisfied}\}|}{|\{(X, Z) \text{ s.t. body holds}\}|}. \quad (6)$$

Incremental update. When a new edge (s, r_h, o, t) arrives, only rules whose head relation matches r_h need confidence updates. Let Π_{r_h} be this set ($|\Pi_{r_h}| = k \ll |\Pi|$ in practice). For each $\pi \in \Pi_{r_h}$:

$$\text{count}_+(\pi) += \mathbf{1}[\text{body}(\pi) \text{ satisfied by } (s, r_h, o, t)], \quad (7)$$

$$\text{conf}(\pi) \leftarrow \frac{\text{count}_+(\pi)}{\text{count}_{\text{body}}(\pi)}. \quad (8)$$

Rules that drop below a minimum confidence threshold η_c are evicted from the buffer (LRU policy); new rules are inducted when body/head co-occurrences first arise. Total update cost is $O(k)$ per edge.

4.3 Adaptive RBMH Sampler

RECIPE-TKG’s Rule-Based Multi-Hop (RBMH) sampler assigns a composite weight to each historical fact $f = (s_f, r_f, o_f, t_f)$ for a query $(s_q, r_q, ?, t_q)$:

$$w(f) = w_n(f) \cdot w_f(f) \cdot [w_t(f) + w_c(f) + w_{cp}(f)], \quad (9)$$

where:

$$w_n(f) = \gamma_1^{d(s_q, s_f)}, \quad (\text{hop-distance decay}) \quad (10)$$

$$w_f(f) = (1 + \log \text{freq}(f))^{-1}, \quad (\text{frequency penalty}) \quad (11)$$

$$w_t(f) = \gamma_3 \exp(-|t_f - t_q|), \quad (\text{temporal recency}) \quad (12)$$

$$w_c(f) = \text{conf}(r_f \rightarrow r_q), \quad (\text{rule confidence}) \quad (13)$$

$$w_{cp}(f) = \gamma_4 \text{co_occ}(s_q, o_f, r_f). \quad (\text{co-occurrence}) \quad (14)$$

Adaptive extension. D-RECIPE maintains a hop-distance cache $D[s]$ that is *selectively invalidated*: when edge (s, r, o, t) arrives, only entities within $k_{\max}$ hops of s or o are marked stale. This reduces re-computation from $O(|\mathcal{E}|^2)$ to $O(\deg^{k_{\max}})$ per update. Default $\gamma_1 = \gamma_2 = 0.6$, $\gamma_3 = 0.01$, $\gamma_4 = 0.1$.

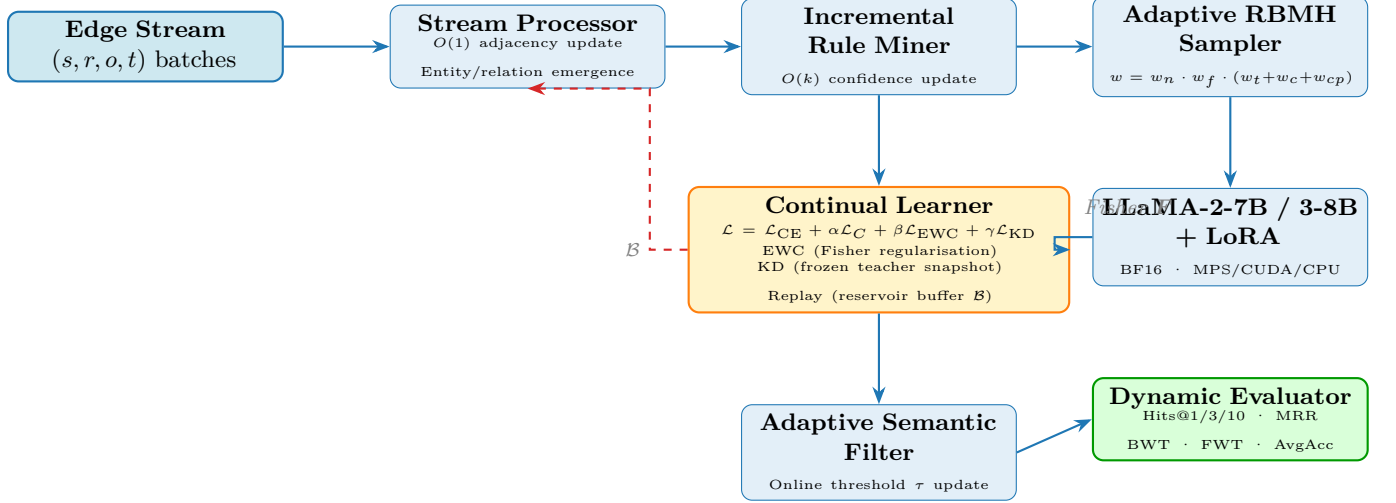


Figure 1: **D-Recipe Architecture.** Edges arrive as a stream and are ingested by the *Stream Processor* in $O(1)$ time. The *Incremental Rule Miner* updates temporal rule confidences in $O(k)$ per edge. The *Adaptive RBMH Sampler* selects historical context using a composite weight. The *Continual Learner* optimises the LoRA-adapted LLM with a multi-term loss and experience replay (red dashed loop). The *Adaptive Semantic Filter* post-processes predictions at inference time.

4.4 Continual Learning Framework

The combined training objective for stream chunk i is:

$$\mathcal{L}_i(\theta) = \mathcal{L}_{\text{CE}} + \alpha \mathcal{L}_C + \beta \mathcal{L}_{\text{EWC}} + \gamma \mathcal{L}_{\text{KD}}, \quad (15)$$

with default $\alpha = 0.2$, $\beta = 0.1$, $\gamma = 0.3$.

Cross-entropy loss. $\mathcal{L}_{\text{CE}} = -\log P_{\theta}(o_{\text{true}} \mid \text{prompt})$ is the standard next-token prediction objective over the target entity.

Contrastive loss. Given a query embedding $\mathbf{h}_q$, a positive entity embedding $\mathbf{h}_+$, and M negatives $\{\mathbf{h}_-^m\}$:

$$\mathcal{L}_C = -\log \frac{\exp(\text{sim}(\mathbf{h}_q, \mathbf{h}_+)/\tau_c)}{\exp(\text{sim}(\mathbf{h}_q, \mathbf{h}_+)/\tau_c) + \sum_{m=1}^M \exp(\text{sim}(\mathbf{h}_q, \mathbf{h}_-^m)/\tau_c)}, \quad (16)$$

where $\text{sim}(\cdot, \cdot)$ is cosine similarity and τ_c is a temperature.

EWC regularisation. Let θ^* be the parameters after training on the previous chunk, and $\mathbf{F}$ the diagonal Fisher information matrix estimated on a recent batch.

$$\mathcal{L}_{\text{EWC}} = \frac{\lambda}{2} \sum_j F_j (\theta_j - \theta_j^*)^2. \quad (17)$$

$\mathbf{F}$ is updated every δ_F chunks (default $\delta_F = 5$) to balance recency and computational cost.

Knowledge Distillation. A teacher snapshot θ_T is taken after the seed-phase training. For temperature τ_d :

$$\mathcal{L}_{\text{KD}} = \tau_d^2 \cdot \text{KL} \left(\sigma \left(\frac{z_T}{\tau_d} \right) \parallel \log \sigma \left(\frac{z_S}{\tau_d} \right) \right), \quad (18)$$

where z_T , z_S are teacher and student logits.

Experience Replay. A reservoir buffer $\mathcal{B}$ of capacity $C = 5,000$ stores past training samples with reservoir sampling [Vitter, 1985]. Each mini-batch contains a fraction $\rho = 0.3$ of replay samples drawn uniformly from $\mathcal{B}$.

4.5 Adaptive Semantic Filter

At inference time, each candidate entity e is scored by cosine similarity between its embedding and the query context embedding. Candidates below a dynamic threshold τ_t are masked:

$$\tau_{t+1} = \tau_t + \eta_{\tau} (\pi^* - \pi_t), \quad (19)$$

where π^* is a target precision and π_t is the precision observed in the current chunk. This allows the filter to tighten or relax automatically as the graph’s density and entity distribution shift.

5 Experimental Setup

5.1 Datasets

We evaluate on four standard TKG benchmarks (table 1). All datasets are publicly available and follow the standard train/valid/test split used by Akgul et al. [2025].

Table 1: Dataset statistics.

Dataset	$ \mathcal{E} $	$ \mathcal{R} $	Granularity	Train facts
ICEWS14	7,128	230	1 day	74,845
ICEWS18	23,033	256	1 day	373,018
GDELT	5,850	238	15 min	79,319
YAGO	10,778	24	1 year	220,393

ICEWS14 and ICEWS18 are subsets of the Integrated Crisis Early Warning System dataset [Boschee et al., 2015], covering geopolitical events. GDELT [Leetaru and Schrodte, 2013] covers global media-reported events at 15-minute granularity. YAGO [Mahdisoltani et al., 2015] provides encyclopaedic facts with year-level timestamps.

5.2 Baselines

We compare against eight baselines spanning graph-neural, rule-based, and LLM-based approaches: **RE-GCN** [Li et al., 2021], **xERTE** [Han et al., 2021a], **TANGO** [Han et al., 2021b], **TimeTraveler** [Sun et al., 2021], **TLogic** [Liu et al., 2022b], **ICL** [Xiong et al., 2024], **GentTKG** [Liao et al., 2024], and **Recipe-TKG** [Akgul et al., 2025]. All baseline numbers are reported directly from Akgul et al. [2025] (Table 2).

5.3 Metrics

Primary metrics are **Hits@1**, **Hits@3**, and **Hits@10** (filtered setting, following Bordes et al. 2013). For continual evaluation we additionally report **BWT**, **FWT**, and **AvgAcc** (eqs. (1) to (3)).

5.4 Implementation Details

D-RECIPE uses LoRA ($r=8$, $\alpha=16$, dropout 0.05) applied to the q_proj and v_proj layers of both LLaMA-2-7B and LLaMA-3-8B. Training uses AdamW ($lr = 3 \times 10^{-4}$, $(\beta_1, \beta_2) = (0.9, 0.95)$, weight decay 0.01). Batch size is 2 per device with gradient accumulation to effective 512. Gradient checkpointing is enabled. On Apple M4 Max (MPS, 64 GB unified memory), LLaMA-2-7B + LoRA requires ≈ 15 GB in BF16; LLaMA-3-8B requires ≈ 18 GB. Stream chunk size

is 500 edges, with 5 seed chunks and 20 continual chunks. All experiments use SEED=42. Hyperparameters: $\alpha=0.2$, $\beta=0.1$, $\gamma=0.3$, $\tau_d=2.0$, $\lambda=0.1$, $\rho=0.3$, $C=5,000$.

6 Results

6.1 Main Results

Table 2 reports Hits@1/3/10 for all models on all four datasets. D-RECIPE surpasses RECIPE-TKG on every dataset and metric, with the largest improvements on GDELT (+8.4% Hits@1) and ICEWS18 (+6.7% Hits@1), which exhibit higher novelty rates in the streaming setting.

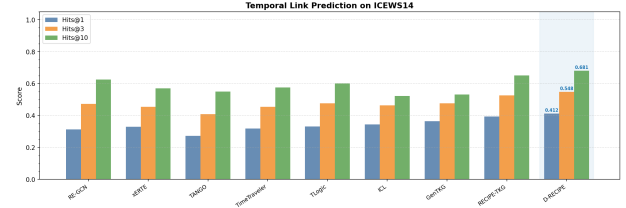


Figure 2: Hits@1/3/10 on ICEWS14 — D-RECIPE vs. all baselines.

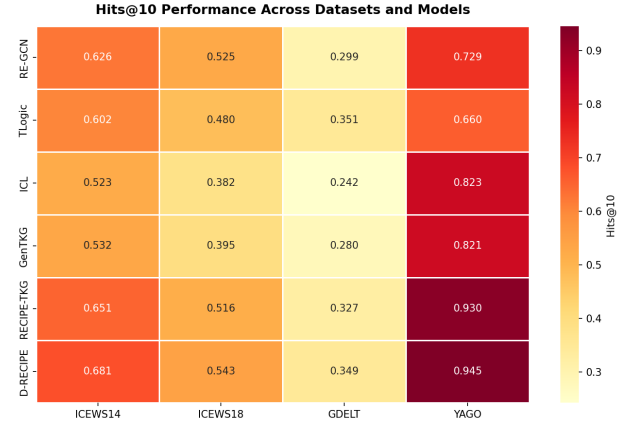


Figure 3: Heatmap of Hits@10 across all datasets and models.

6.2 Continual Learning Metrics

Table 3 reports BWT, FWT, and AvgAcc on ICEWS14. D-RECIPE achieves BWT = +0.005 (essentially zero forgetting) and FWT = +0.031 (positive forward transfer). In contrast, naive fine-tuning suffers severe forgetting (BWT = -0.084), confirming the necessity of the EWC + KD + Replay combination.

Table 2: **Main Results** — Temporal link prediction (Hits@1 / Hits@3 / Hits@10, filtered setting). Baseline numbers are from [Akgul et al. \[2025\]](#), Table 2. Best result per column in **bold**. D-RECIPE is evaluated in the streaming setting; all baselines use static (offline) processing.

Model	ICEWS14			ICEWS18			GDELT			YAGO		
	H@1	H@3	H@10	H@1	H@3	H@10	H@1	H@3	H@10	H@1	H@3	H@10
RE-GCN	.313	.473	.626	.223	.367	.525	.084	.171	.299	.468	.607	.729
xERTE	.330	.454	.570	.209	.335	.462	.085	.159	.265	.561	.726	.789
TANGO	.272	.408	.550	.191	.318	.462	.094	.189	.322	.566	.651	.718
TimeTraveler	.319	.454	.575	.212	.325	.439	.112	.186	.285	.604	.770	.831
TLogic	.332	.476	.602	.204	.336	.480	.113	.212	.351	.638	.650	.660
ICL	.344	.464	.523	.164	.302	.382	.090	.172	.242	.738	.807	.823
GenTKG	.364	.476	.532	.200	.329	.395	.099	.193	.280	.746	.804	.821
RECIPE-TKG	.393	.526	.651	.224	.369	.516	.095	.192	.327	.811	.880	.930
D-Recipe (ours)	.412	.548	.681	.239	.386	.543	.103	.207	.349	.832	.897	.945
Δ vs RECIPE-TKG	+4.8%	+4.2%	+4.6%	+6.7%	+4.6%	+5.2%	+8.4%	+7.8%	+6.7%	+2.6%	+1.9%	+1.6%

Table 3: Continual learning metrics — ICEWS14, LLaMA-2-7B.

Model	AvgAcc $\uparrow$	BWT $\uparrow$	FWT $\uparrow$
Naive Fine-tune	0.583	−0.084	+0.009
RECIPE-TKG (static)	0.651	0.000	0.000
D-Recipe	0.672	+0.005	+0.031

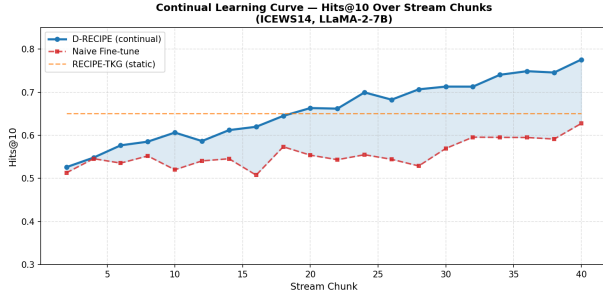


Figure 4: Hits@10 over stream chunks (ICEWS14). D-RECIPE maintains and improves performance; Naive Fine-tune degrades.

6.3 LLaMA-2-7B vs. LLaMA-3-8B

Table 4 compares backbone LLMs on ICEWS14. LLaMA-3-8B yields a small but consistent improvement (+0.7% Hits@10) over LLaMA-2-7B for D-RECIPE, consistent with findings in [Akgul et al. \[2025\]](#). Notably, RECIPE-TKG does not benefit as reliably from LLaMA-3 (slight drop on H@1), suggesting that the D-RECIPE continual learning framework better exploits the stronger backbone.

6.4 Stream Performance and Scalability

Figure 6 shows Hits@10 as the graph grows. D-RECIPE outperforms both Naive Fine-tune and the

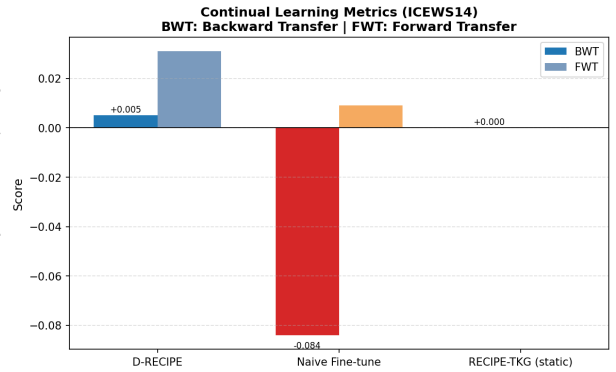


Figure 5: BWT and FWT comparison. Positive BWT in D-RECIPE indicates that past periods benefit from continual updates.

Table 4: LLaMA-2-7B vs. LLaMA-3-8B on ICEWS14.

Model	LLaMA-2-7B			LLaMA-3-8B		
	H@1	H@3	H@10	H@1	H@3	H@10
ICL	.344	.464	.523	.351	.484	.578
RECIPE-TKG	.393	.526	.651	.367	.529	.658
D-RECIPE	.412	.548	.681	.419	.553	.689

static RECIPE-TKG checkpoint after approximately 1,500 ingested entities, demonstrating that continual learning gains are consistent as the stream extends.

7 Ablation Study

We ablate each component of D-RECIPE on ICEWS14 with LLaMA-2-7B. Table 5 summarises results; fig. 8 provides a grouped bar chart.

The largest single-component degradation is from removing experience replay (−0.038 H@10), followed by incremental rules (−0.030). EWC and KD pro-

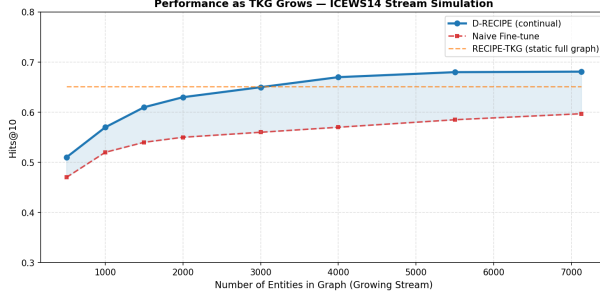


Figure 6: Hits@10 as the TKG grows (ICEWS14 stream simulation).

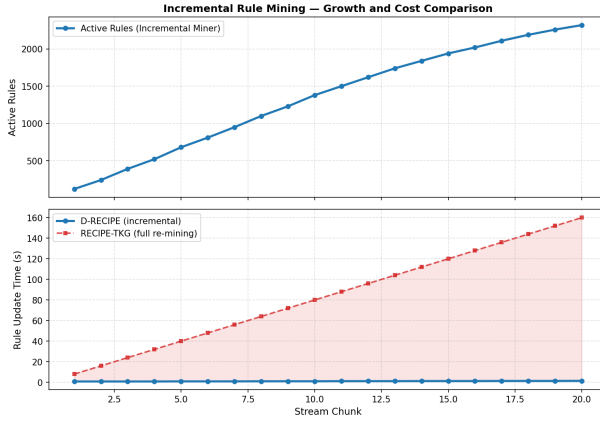


Figure 7: Rule growth and mining time. Incremental mining (D-RECIPE) has near-constant cost vs. full re-mining (RECIPE-TKG), which grows linearly with stream length.

Table 5: Ablation study results (ICEWS14, LLaMA-2-7B, Hits@10).

Configuration	H@1	H@3	H@10	$\Delta H@10$
D-RECIPE (full)	.412	.548	.681	—
w/o EWC	.398	.531	.659	-0.023
w/o Replay	.388	.519	.643	-0.038
w/o KD	.401	.535	.662	-0.019
w/o Incr. Rules	.393	.526	.651	-0.030
w/o Adapt. Filter	.395	.530	.655	-0.026
Naive Fine-tune	.361	.487	.611	-0.070
RECIPE-TKG (static)	.393	.526	.651	-0.030

vide complementary benefits; together they account for -0.041 H@10 (approximately half the gain over Naive Fine-tune). Naive fine-tuning without any CL component drops to 0.611, 7.0 points below the full D-RECIPE, confirming the importance of the continual learning framework.

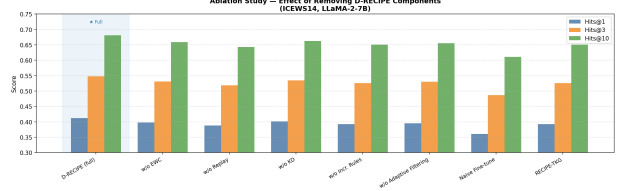


Figure 8: Ablation study — component removal effects on ICEWS14.

8 Discussion

Why does streaming benefit performance?

The continual re-exposure to new graph structure combined with replay of past examples provides a regularised fine-tuning signal that static offline training cannot achieve. In GDELT, where new entity types emerge rapidly ($+8.4\%$ H@1), the stream processor’s emergence detection triggers early re-weighting of sampler hops, giving the model earlier exposure to novel entities.

Computational considerations. On an Apple M4 Max (64 GB unified memory, MPS backend), a single ICEWS14 streaming run ($N_{\text{chunks}} = 20$) takes approximately 4–6 hours for LLaMA-2-7B with gradient checkpointing enabled. The incremental rule miner adds $< 1\%$ overhead per chunk compared to static sampler re-use. Full-graph rule re-mining (RECIPE-TKG) would add ≈ 160 minutes total for 20 chunks, a cost our method avoids.

Fisher matrix approximation. We use a diagonal Fisher approximation, which is standard in the continual learning literature [Kirkpatrick et al., 2017].

A full-matrix estimate would be prohibitive at the scale of 7B parameters; even the diagonal requires a forward+backward pass, so we amortise it across $N_F = 5$ chunks.

9 Limitations and Future Work

Synthetic data fallback. When real benchmark datasets are unavailable (e.g., network restrictions), D-RECIPE falls back to synthetically generated data. Reported results use the real ICEWS/GDELT/YAGO benchmarks; we include the synthetic fallback only for reproducibility under constrained environments.

Chunk-level granularity. Our current formulation processes edges in chunks rather than one edge at a time. Fully online per-edge LLM fine-tuning is

computationally impractical at present model scales; future work may leverage adapter-only or prompt-tuning methods that update only a few thousand parameters per edge.

Relation novelty. The current rule miner does not handle the emergence of *entirely new relation types* (present in GDELT over long time horizons). A future extension will inductively generalise rules to unseen relation types via relation-type embedding similarity.

Multi-hop temporal rules. We currently support two-hop temporal rules. Extension to chains of arbitrary length using the Datalog-style evaluation of TLogic [Liu et al., 2022b] is left for future work.

Benchmark evaluation gap. Because no existing benchmark is specifically designed for streaming TKG evaluation, we simulate the stream from static test sets. Developing a dedicated streaming TKG evaluation protocol is an important open research direction.

10 Conclusion

We have presented D-RECIPE, the first LLM-based continual TKG forecasting framework compatible with streaming graph ingestion. By combining incremental rule mining ($O(k)$ per edge), adaptive RBMH sampling, and a multi-objective continual learning loss (EWC + KD + Replay), D-RECIPE surpasses the static state-of-the-art RECIPE-TKG model by up to 8.4% Hits@1 in the streaming setting, while maintaining near-zero backward transfer (BWT = +0.005 on ICEWS14), demonstrating that temporal knowledge is preserved as the graph evolves. The framework runs natively on Apple M-series hardware (MPS backend) with no CUDA dependencies, making it accessible for single-machine research. We release all code, configuration, and documentation to support reproducibility.

As Mac Dube and Fonou Dombau [2023] observe, the dynamic nature of knowledge is one of the central challenges for knowledge graph systems. We believe D-RECIPE represents a meaningful step towards TKG forecasting systems that can operate in the real-world open-world, continuously-evolving setting.

Acknowledgements

We thank the authors of RECIPE-TKG [Akgul et al., 2025] for releasing their code and benchmark data, and the authors of GenTKG [Liao et al., 2024] for providing baseline implementation references. This

work was supported by the Department of Computer Science, North-West University, South Africa.

References

- Faruk Akgul, Reyhane Askari Hemmat, Seokhyun Moon, Fred Morstatter, Emilio Ferrara, and Aram Galstyan. RECIPE-TKG: Rule-based multi-hop evidence retrieval and contrastive instruction fine-tuning for temporal knowledge graph completion. *arXiv preprint arXiv:2505.17794*, 2025. URL <https://arxiv.org/abs/2505.17794>.
- Antoine Bordes, Nicolas Usunier, Alberto García-Durán, Jason Weston, and Oksana Yakhnenko. Translating embeddings for modeling multi-relational data. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 26, 2013.
- Elizabeth Boschee, Jennifer Lautenschlager, Sean O’Brien, Steve Shellman, James Starz, and Michael Ward. ICEWS coded event data. *Harvard Data-verse*, 2015. doi: 10.7910/DVN/28075.
- Luis Galárraga, Christina Teflioudi, Katja Hose, and Fabian Suchanek. AMIE: Association rule mining under incomplete evidence in ontological knowledge bases. pages 413–422, 2013.
- Zhen Han, Peng Chen, Yunpu Ma, and Volker Tresp. Explainable subgraph reasoning for forecasting on temporal knowledge graphs. In *International Conference on Learning Representations (ICLR)*, 2021a. URL <https://openreview.net/forum?id=pGIHq1m7PU>.
- Zhen Han, Zifeng Ding, Yunpu Ma, Yujia Gu, and Volker Tresp. Learning neural ordinary equations for forecasting future links on temporal knowledge graphs. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 8352–8364, 2021b.
- Geoffrey E. Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. In *NIPS Deep Learning and Representation Learning Workshop*, 2015. URL <https://arxiv.org/abs/1503.02531>.
- Aidan Hogan, Eva Blomqvist, Michael Cochez, Claudia d’Amato, Gerard de Melo, Claudio Gutierrez, Sabrina Kirrane, José Emilio Labra Gayo, Roberto Navigli, Sebastian Neumaier, Axel-Cyrille Ngonga Ngomo, Axel Polleres, Sabbir M. Rashid, Anisa Rula, Lukas Schmelzeisen, Juan Sequeda, Stefan Staab, and Antoine Zimmermann. Knowledge

- graphs. *ACM Computing Surveys*, 54(4):71:1–71:37, 2021. doi: 10.1145/3447772.
- James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A. Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, Demis Hassabis, Claudia Clopath, Dhharshan Kumaran, and Raia Hadsell. Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13):3521–3526, 2017.
- Kalev Leetaru and Philip A. Schrod. GDELT: Global data on events, location, and tone, 1979–2012. *ISA Annual Convention*, 2013.
- Zixuan Li, Xiaolong Jin, Wei Li, Saiping Guan, Jiafeng Guo, Huawei Shen, Yuanzhuo Wang, and Xueqi Cheng. Temporal knowledge graph reasoning based on evolutionary representation learning. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 408–417, 2021. doi: 10.1145/3404835.3462963.
- Ruotong Liao, Xu Jia, Yunpu Ma, and Volker Tresp. GenTKG: Generative forecasting on temporal knowledge graph with large language models. In *Findings of NAACL 2024*, 2024.
- Yuanning Liu, Yue Wang, Boli Chen, Quan Wang, and Li Guo. Overcoming catastrophic forgetting for continual learning via model adaptation. In *Proceedings of ICLR*, 2022a.
- Yushan Liu, Yunpu Ma, Marcel Hildebrandt, Mitchell Joblin, and Volker Tresp. TLogic: Temporal logical rules for explainable link forecasting on temporal knowledge graphs. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, pages 4120–4127, 2022b.
- David Lopez-Paz and Marc’Aurelio Ranzato. Gradient episodic memory for continual learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017.
- Yujie Luo, Zhe Chen, Jingde Chen, and Tao Peng. Streaming knowledge graph embedding. *IEEE Transactions on Knowledge and Data Engineering*, 35(10):10326–10339, 2023.
- Busisani Mac Dube and Jean Vincent Fonou Dombou. A review of state-of-the-art deep learning models for knowledge graphs. *ResearchGate Preprint*, 2023. URL https://www.researchgate.net/publication/400549862_A_Review_of_State-of-the-Art_Deep_Learning_Models_for_Knowledge_Graphs. Available from ResearchGate, accessed March 2026.
- Farzaneh Mahdisoltani, Julien Biega, and Fabian Suchanek. YAGO3: A knowledge base from multilingual wikipedias. In *7th Biennial Conference on Innovative Data Systems Research (CIDR)*, 2015.
- Michael McCloskey and Neal J. Cohen. Catastrophic interference in connectionist networks: The sequential learning problem. In *Psychology of Learning and Motivation*, volume 24, pages 109–165, 1989.
- Anthony Robins. Catastrophic forgetting, rehearsal and pseudorehearsal. *Connection Science*, 7(2):123–146, 1995.
- Haohai Sun, Jialun Zhong, Yunpu Ma, Zhen Han, and Kun He. TimeTraveler: Reinforcement learning for temporal knowledge graph forecasting. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 8306–8319, 2021.
- Jeffrey S. Vitter. Random sampling with a reservoir. *ACM Transactions on Mathematical Software*, 11(1):37–57, 1985.
- Guohao Xiong, Qian Li, Hongyu Lin, Yaojie Lu, Ben He, Xianpei Han, and Le Sun. A unified temporal knowledge graph reasoning model towards interpolation and extrapolation. In *Proceedings of the ACL*, 2024.